

PREDICTING SILVER PRICE TRENDS USING AI AND TIME SERIES MODELS

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ABSTRACT

The dynamic and volatile nature of commodity markets makes accurate price forecasting a critical challenge, particularly in the case of precious metals like silver. This study presents a comprehensive analysis of various forecasting models applied to predict silver price movements in the Indian commodity market. The research integrates traditional time series models—ARIMA, SARIMA, and Holt-Winters Exponential Smoothing (HWES)—with advanced machine learning and deep learning techniques, including Random Forest, Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Convolutional Neural Networks (CNN). Additionally, a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is employed to model the volatility inherent in silver price series, especially through the residuals of ARIMA forecasts.

Daily historical silver price data from 2010 to 2025 was collected and pre-processed for model development using Python libraries such as Pandas, NumPy, Keras, and TensorFlow. Each model's performance was evaluated using standard accuracy metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). The results reveal that machine learning and deep learning models significantly outperform traditional models in predictive accuracy, with Random Forest and LSTM leading the performance charts. The GARCH model further contributes by quantifying volatility patterns, validating the presence of volatility clustering and persistence. The study's findings emphasize the potential of hybrid modeling approaches for enhanced forecasting in financial time series analysis. This paper contributes to the literature by offering a comparative, data-driven perspective on forecasting techniques for silver, aiding investors, traders, and policymakers in strategic decision-making. Future research could extend the framework by integrating macroeconomic variables, applying attention-based models, or incorporating high-frequency trading data for real-time predictions.

Keywords

Silver price forecasting, ARIMA, LSTM, Random Forest, GARCH, Deep Learning, Time Series Models, Volatility Modeling, Indian Commodity Market, Financial Forecasting.

INTRODUCTION

The price of silver has historically held great importance in the Indian economy, functioning as both a valuable investment asset and a culturally significant commodity. Silver has become more volatile in recent years, influenced by global economic instability, varying demand from industrial and jewellery markets, and shifts in currency and commodity prices. For investors, traders, and policymakers, correctly predicting silver price trends is crucial for making decisions, hedging risks, and managing portfolios. Conventional techniques like ARIMA, SARIMA, and Holt-Winters have been extensively applied in forecasting financial time series. These models excel at identifying linear trends and seasonal patterns but struggle with abrupt spikes, volatility, and non-linear market behaviours. The Indian silver market, shaped by local celebrations, global commerce, currency shifts, and economic updates, requires advanced techniques to analyse trends and provide precise forecasts.

The Indian silver market, shaped by local celebrations, global commerce, currency shifts, and economic updates, requires advanced techniques to analyse trends and provide precise forecasts. In this context, Artificial Intelligence (AI) methods—particularly machine learning and deep learning—have demonstrated potential in analysing intricate and noisy financial data. Algorithms such as Random Forest (RF) and Artificial Neural Networks (ANN) can learn from extensive datasets that have numerous characteristics. Deep learning models that are specifically made to handle sequential data, like Gated Recurrent Units (GRU) and Long Short-Term Memory (LSTM), are ideal for financial time series. Despite being created initially for image processing, Convolutional Neural Networks (CNN) have also been successful in time series forecasting through the identification of local patterns.

Through a comparison of their efficacy in predicting silver prices in the Indian commodity market, this study seeks to close the gap between conventional time series models and AI-driven approaches. In addition to identifying the most accurate model, the objective is to determine which approach provides the best balance between interpretability, complexity, and accuracy. Through the application of several models to historical silver price data and their assessment using robust error measures (MAE, RMSE, MSE, and MAPE), this study offers a thorough comparison to determine the best accurate forecasting method. This study is useful for traders, investors, financial analysts, and even politicians who want data-driven insights

into the behaviour of the silver price, as predictive analytics is becoming more and more significant in commodities markets. This contributes to the increasing amount of research that supports the use of intelligent and hybrid systems for financial forecasting.

BACKGROUND OF THE STUDY

Because it is classified as both an industrial and a precious metal, silver is a key component of both the global and Indian commodity markets. In addition to its considerable usage in electronics, photovoltaics, and medicine, silver has historically been appreciated for its beauty and investment qualities. The demand for silver in India is driven by both global price fluctuations and indigenous sociocultural elements (such as weddings and festivals), which makes its pricing behaviour extremely complex and dynamic. Time series models based on the statistical heritage of linear forecasting, such as ARIMA and Holt-Winters, have historically been used extensively in silver price forecasting (Box & Jenkins, 1976). Under steady trend and seasonality settings, these models perform well and are interpretable. They are inadequate, nonetheless, in the face of market shocks, structural cracks, and non-linearity.

Traditional models found it difficult to account for the sudden market volatility brought about by the COVID-19 epidemic and geopolitical tensions like the Russia-Ukraine war (Alam & Rath, 2021). To get beyond these restrictions, academics have been looking more closely at what machine learning (ML) and deep learning (DL) can do in recent years. By identifying non-linear connections and interactions across several variables, machine learning models like Random Forest, Support Vector Regression, and Boost have proven to perform better in forecasting (Zhou et al., 2020). Due to their capacity to learn temporal dependencies and preserve long-term memory, deep learning models such as LSTM and GRU have shown exceptional efficacy for time series (Ahmed et al., 2022). Due to their ability to achieve high accuracy in complicated contexts, hybrid models—which integrate traditional and AI techniques—are becoming more and more popular (Singh & Yadav, 2023).

The superiority of AI models over conventional methods in predicting not just stock prices but also the values of commodities like gold, crude oil, and silver has been confirmed by recent studies. Both scholarly study and real-world applications have changed because of the transition from linear models to intelligent forecasting systems. AI-driven techniques are now practical and beneficial for real-time commodity market analysis because to the expansion of high-frequency financial data availability and the rise in processing capacity (Kumar & Jain, 2024).

PROBLEM STATEMENT

Predicting silver prices in India is getting harder in the quickly changing economic climate of today. In contrast to earlier times when straightforward historical patterns might provide reasonable predictions, a variety of intricate factors—including global inflation, industrial demand, exchange rates, geopolitical developments, and even shifting investor sentiment—have an impact on the silver market today. Though effective in identifying patterns and seasonality, traditional forecasting models such as ARIMA and Holt-Winters frequently fail to capture the whole picture, particularly in the event of abrupt spikes or non-linear shifts.

In addition, new technologies like deep learning and machine learning are enabling the extraction of patterns from big, noisy datasets. Although models such as LSTM and Random Forest have demonstrated potential in other financial fields, little is known about their precise use and relative success in the Indian silver market. Policy analysts, traders, and investors are left wondering which model they can rely on to produce the most accurate forecasts. And at what price in reliability, interpretability, and complexity?

That fundamental problem—how can we more precisely predict silver price changes in a market that no longer exhibits predictable behaviour—is the driving force behind this study. To ascertain not only which model performs best, but also why and under what circumstances, it is evident that both conventional and AI-driven techniques need to be assessed and contrasted. Since silver is a significant asset class in Indian homes, investments, and industry, finding a solution to this issue is both urgent and useful in the real world.

NEED FOR THE STUDY

In the cultural and economic context of India, silver occupies a special place. A vital industrial component for industries like electronics, solar energy, and pharmaceuticals, it is also a valuable metal used in jewellery and traditional ceremonies. But in recent years, currency fluctuations, commodities speculation, geopolitical events, and changes in the world economy have all had an impact on the price of silver, making it increasingly volatile. Accurate silver price forecasting is crucial in such an unpredictable climate for traders, investors, financial institutions, and even regulators.

Although conventional time series models, such as ARIMA and SARIMA, have been extensively employed for commodity price forecasting, they frequently fall short in identifying non-linear trends, abrupt shocks, or market structural shifts. However, sophisticated AI-based

models like deep learning and machine learning have shown impressive results in financial forecasting, especially when it comes to identifying intricate patterns and long-term connections. Nevertheless, there aren't many thorough studies that examine these models specifically for predicting the price of silver in the Indian setting. Assessing the forecasting methods' accuracy, consistency, and suitability with real-time silver price data is becoming increasingly important.

Testing whether these more recent models genuinely provide appreciable gains in forecasting accuracy over previously established ones is pertinent and timely as AI tools become more widely available and computational resources become more reasonably priced. Furthermore, choosing the best model will help market players as well as create trading and investment techniques that are more resilient. To close the gap between new intelligent systems and traditional statistical forecasting, this work is crucial. The study will give stakeholders in India's financial and commodity markets useful information and a framework for making decisions by comparing the two model types on a shared platform using accepted evaluation metrics.

OBJECTIVES

1. To analyse historical silver price trends using traditional time series models such as ARIMA, SARIMA, and Holt-Winters.
2. To apply machine learning and deep learning models—including Random Forest (RF), Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Convolutional Neural Networks (CNN)—for forecasting silver prices.
3. To evaluate and compare the forecasting performance of traditional, machine learning, and deep learning models using standard error metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE).
4. To identify the most suitable forecasting model for silver prices in the Indian commodity market based on accuracy, robustness, and practical usability.

LITERATURE REVIEW

1. **Kumar and Jain (2024)**, "*Intelligent systems for real-time commodity price forecasting: A machine learning perspective*". The study sought to determine how well machine

learning algorithms predicted current commodity prices, with a particular emphasis on crude oil and silver. Based on past pricing data and external macroeconomic indicators, the researchers developed prediction models using methods like Random Forest, XGBoost, and Support Vector Regression (SVR). The results showed that, particularly during times of market volatility, ensemble models such as XGBoost consistently performed better than conventional statistical models. The authors concluded that intelligent systems offer more accurate, flexible, and timely predictions for commodity prices in developing markets such as India.

2. **Singh and Yadav (2023)**, *“Hybrid forecasting models: A review of ARIMA and deep learning integration in financial markets”*. This study sought to investigate the possibilities of hybrid forecasting models for financial and commodities market prediction, which combine deep learning models like LSTM and GRU with conventional time series techniques like ARIMA. Using historical data from the metal and energy markets, such as those for crude oil and silver, the authors examined and tried hybrid combinations. The process comprised breaking down the time series using ARIMA to model the linear component and then using LSTM to the residuals to account for non-linear behavior. The results indicated that hybrid models performed much better than standalone models in terms of MAPE and RMSE, which makes them perfect for financial data that is non-stationery and erratic.
3. **Sharma and Raj (2022)**, *“Machine learning-based ensemble approaches for forecasting silver prices in India”*. This study aimed to develop and compare ensemble machine learning models for forecasting silver prices in the Indian commodity market. The authors applied models such as Random Forest, Gradient Boosting, and XGBoost using a dataset of daily silver prices and relevant macroeconomic indicators like the USD-INR exchange rate and gold prices. The models were evaluated using MSE, RMSE, and MAPE. The results indicated that XGBoost provided the most accurate forecasts, especially during periods of high market volatility. The study concluded that ensemble learning methods outperform traditional statistical models and offer practical value for real-world forecasting applications in commodity trading.
4. **Ahmed, Li, and Zhang (2022)**, *“Forecasting commodity prices using LSTM and GRU: A case of precious metals”*. This paper aimed to assess the forecasting accuracy of deep

learning models—specifically LSTM and GRU—in predicting the prices of precious metals such as gold and silver. Using daily closing prices from international markets, the authors trained both models on sequential datasets and evaluated performance using MAE, RMSE, and MAPE. The methods focused on capturing temporal dependencies and non-linear trends within the price data. The findings revealed that GRU slightly outperformed LSTM in terms of speed and accuracy, particularly for shorter forecasting horizons. The study emphasized the relevance of deep learning models for high-frequency financial data with complex structures.

5. **Pal, Ghosh, and Bera (2021)**, “*Comparative study of ARIMA, SVR and LSTM models in predicting precious metal prices*”. This research aimed to compare the predictive performance of ARIMA, Support Vector Regression (SVR), and Long Short-Term Memory (LSTM) models in forecasting precious metal prices, with a focus on gold and silver. The models were trained using historical daily price data from international markets and evaluated through MAE and RMSE metrics. ARIMA was used as a baseline, while SVR captured non-linear patterns and LSTM modelled sequential dependencies. The results showed that LSTM delivered the highest accuracy, particularly in long-term forecasts, whereas ARIMA performed well only under stable market conditions. The study concluded that deep learning models are more robust for volatile commodity markets.
6. **Alam and Rath (2021)**, “*The impact of COVID-19 on commodity markets: A case of silver and gold*”. This study aimed to investigate how the COVID-19 pandemic affected the pricing and volatility of precious metals, specifically silver and gold. The authors employed GARCH family models to analyse volatility clustering and price behavior using daily data from January 2019 to December 2020. The study also incorporated structural break tests to capture the impact of key pandemic-related events. The findings revealed a significant increase in silver price volatility during the early phases of the pandemic, with GARCH models capturing this better than ARIMA. The paper emphasized the limitations of traditional models in times of global uncertainty and advocated for volatility-aware forecasting approaches.
7. **Zhou, Wang, and Luo (2020)**, “*A machine learning framework for commodity price prediction: Case studies on energy and metals*”. This study aimed to build a generalizable machine learning framework for predicting commodity prices, including metals like silver and gold. The researchers used a combination of algorithms—Random Forest, XGBoost,

and Support Vector Machines—trained on a multi-dimensional dataset consisting of historical prices, exchange rates, and macroeconomic indicators. The models were evaluated using cross-validation and standard metrics like RMSE and MAPE. Results showed that XGBoost consistently achieved the lowest error rates across all commodities, outperforming both traditional time series models and other machine learning techniques. The study concluded that machine learning, particularly ensemble methods, offers scalable and accurate forecasting tools for dynamic markets.

8. **Jain and Chatterjee (2020)**, “Comparative analysis of LSTM and ARIMA models in predicting precious metal prices”. The objective of this study was to compare the effectiveness of Long Short-Term Memory (LSTM) networks and ARIMA models in forecasting precious metal prices, with a particular focus on silver. The researchers used daily spot price data and trained both models under identical conditions to ensure fairness in comparison. Evaluation metrics included MAE, RMSE, and MAPE. The LSTM model showed significantly better accuracy, especially during volatile market phases, whereas ARIMA failed to adapt to sharp price swings. The study concluded that LSTM's ability to capture long-term dependencies and non-linearity made it a more suitable option for silver price forecasting.
9. **Patel and Desai (2019)**, “*Application of machine learning algorithms for commodity price forecasting: A case of silver*”. This study aimed to evaluate the performance of various machine learning algorithms—namely Decision Trees, Random Forest, and Support Vector Regression—in forecasting silver prices using historical data from the Indian market. The authors also incorporated external factors such as global commodity indices and exchange rate trends. After preprocessing the data and training the models, performance was assessed using RMSE and MAPE. The results indicated that Random Forest delivered the most consistent and accurate forecasts among the tested models. The study concluded that machine learning techniques, particularly ensemble methods, offer considerable advantages over traditional models in predicting silver prices with complex influences.
10. **Verma and Mehta (2019)**, “Time series forecasting of silver prices using ARIMA and Holt-Winters models”. The study aimed to forecast silver prices in the Indian market using two traditional time series models—ARIMA and Holt-Winters exponential smoothing. Daily price data over a five-year period was used, and both models were evaluated based

on AIC, RMSE, and MAPE values. The authors found that the Holt-Winters model performed slightly better during seasonal fluctuations, while ARIMA was more suited for short-term linear trends. However, both models exhibited limitations in capturing sudden price spikes. The study highlighted the need for more adaptive models that could incorporate exogenous variables and respond to real-time volatility in the silver market.

11. Bansal and Kumar (2018), “Volatility modeling and forecasting of silver prices using GARCH family models”. This study aimed to understand and forecast the volatility of silver prices using GARCH, EGARCH, and GJR-GARCH models. The researchers used daily silver price data from the Multi Commodity Exchange (MCX) and applied the models to detect time-varying volatility patterns and asymmetric responses to market news. The models were evaluated using AIC, BIC, and out-of-sample forecast accuracy. Findings showed that EGARCH performed best in capturing volatility clustering and leverage effects, making it suitable for forecasting during turbulent periods. The study emphasized that volatility-based models are more effective than linear models in risk-sensitive financial forecasting scenarios.

12. Roy and Sengupta (2018), “Forecasting precious metal prices using hybrid ARIMA-ANN model”. This study aimed to develop a hybrid forecasting model by combining ARIMA for linear components and Artificial Neural Networks (ANN) for non-linear residuals to improve the accuracy of silver and gold price predictions. The authors used daily international price data over a six-year period and applied ARIMA first to model the trend, followed by ANN on the residuals. Performance was evaluated using RMSE, MAPE, and Theil’s U-statistic. The hybrid ARIMA-ANN model outperformed both standalone ARIMA and ANN models, particularly during volatile market phases. The study concluded that hybrid models are effective in capturing both predictable and irregular price movements in precious metals.

13. Tripathi and Ghosh (2017), “Forecasting metal prices using neural networks and wavelet transforms”. This study aimed to forecast metal prices, including silver, by combining wavelet decomposition with artificial neural networks (ANN). The authors decomposed the price time series into multiple frequency components using discrete wavelet transform and then trained ANN on each component. This hybrid approach aimed to improve the model’s ability to detect short-term spikes and long-term trends. The results demonstrated superior

performance over standalone ANN and ARIMA models, with lower RMSE and MAPE values. The study concluded that wavelet-ANN hybrids are highly effective for volatile commodity forecasting due to their ability to capture multi-scale patterns.

- 14. Reddy and Thomas (2016),** *“ARIMA and ARCH modeling for silver price analysis in Indian markets”*. The objective of this study was to compare ARIMA and ARCH models in capturing the behavior of silver prices in Indian financial markets. The authors used daily data from the MCX and conducted model diagnostics including Ljung-Box tests and ARCH effects. ARIMA was found to be suitable for short-term forecasts, while ARCH captured volatility better. Combined use of both models offered moderate forecasting accuracy. The study highlighted the limitations of traditional linear models and the importance of accounting for conditional variance in financial time series.
- 15. Jaiswal and Banerjee (2016),** *“A comparative study of SVR and ANN in commodity price prediction”*. This paper aimed to evaluate the effectiveness of Support Vector Regression (SVR) and Artificial Neural Networks (ANN) in forecasting commodity prices, with a focus on silver and crude oil. Both models were trained using technical indicators and historical price data. Results indicated that SVR had better generalization and lower prediction error during stable market conditions, while ANN performed better in volatile periods. The study concluded that model selection should depend on market phase and suggested combining both approaches for improved accuracy.
- 16. Bhattacharya and Sinha (2015),** *“Short-term forecasting of silver and gold prices using Holt-Winters and exponential smoothing techniques”*. This study focused on short-term forecasting using Holt-Winters and single exponential smoothing models. Daily data was collected for silver and gold prices from international markets, and models were evaluated based on forecast error and responsiveness to recent changes. While both models showed acceptable performance in detecting seasonal trends, they failed to adapt quickly during sharp price reversals. The authors concluded that although exponential smoothing is useful for quick trend-based forecasting, it should be supplemented by more adaptive models in high-volatility periods.
- 17. Prasad and Iyer (2015),** *“Using ARIMA-GARCH models for financial time series forecasting: A case of silver prices”*. The authors aimed to build a composite ARIMA-

GARCH model to forecast silver price levels and their volatility. ARIMA was used to model the mean process, while GARCH modelled the variance. The dataset included international silver price data over ten years. The combined model outperformed standalone ARIMA and GARCH in forecasting performance, particularly during periods of high market uncertainty. The study advocated the use of integrated modeling frameworks for both level and volatility forecasting in commodity markets.

- 18. Meena and Khanna (2014),** *“Forecasting precious metal prices with neural networks: An Indian perspective”*. This paper aimed to explore the capability of feedforward neural networks in forecasting the prices of precious metals including silver in India. Daily price data was normalized and fed into a multi-layer perceptron model. The results were compared with linear regression and ARIMA models. Neural networks demonstrated superior forecasting accuracy with a noticeable reduction in MAPE, especially during festival and budget announcement periods. The authors concluded that neural networks could adapt well to the unpredictable nature of Indian commodity markets.
- 19. Sahu and Ranjan (2013),** *“Comparative study of ARIMA and multiple linear regression for forecasting silver prices”*. The objective of this study was to compare the effectiveness of ARIMA and Multiple Linear Regression (MLR) models in forecasting silver prices using historical price data and macroeconomic indicators such as inflation rate and USD-INR exchange rate. The authors found that ARIMA models provided better short-term forecasts, while MLR was more useful in interpreting economic relationships. However, both models suffered from limitations in handling volatility. The study called for more robust models capable of capturing structural breaks and sudden shocks.
- 20. Sharma and Dey (2012),** *“Modeling and forecasting silver prices in India using statistical time series techniques”*. This foundational study aimed to model and forecast silver prices in India using classical time series techniques including ARIMA, exponential smoothing, and moving average models. The research utilized ten years of monthly silver price data and focused on forecasting for investment purposes. While ARIMA provided the best fit among the tested models, it struggled during periods of abrupt price changes. The study concluded that while time-tested statistical models are useful for baseline forecasts, they are insufficient for handling modern-day market complexities without external variable inputs or adaptive mechanisms.

METHODS AND MEASURES

The aim of this study is to forecast the daily prices of silver in the Indian commodity market using a combination of traditional time series models, machine learning techniques, and deep learning frameworks. The selected models include **ARIMA**, **SARIMA**, **Holt-Winters Exponential Smoothing (HWES)**, **Random Forest**, **Artificial Neural Networks (ANN)**, **Long Short-Term Memory (LSTM)**, **Gated Recurrent Units (GRU)**, **Convolutional Neural Networks (CNN)**, and a **hybrid ensemble model**. These models are implemented using Python programming with support from libraries such as **pandas**, **NumPy**, **stats models**, **scikit-learn**, **Keras**, and **TensorFlow**.

Data Collection and Preprocessing

The dataset comprises daily silver prices collected from reliable financial sources such as **MCX India**, **Investing.com**, and **Yahoo Finance**, covering the period from **January 1, 2010, to June 30, 2025**. The data undergoes preprocessing to address missing values, remove outliers, normalize prices, and convert the time series into supervised learning formats for ML and DL models. Following **Kumar & Gupta (2021)**, the LSTM and CNN models also incorporate **technical indicators** such as **Moving Averages (MA)** and **Relative Strength Index (RSI)** to enhance predictive accuracy.

Traditional Models

- **ARIMA (Autoregressive Integrated Moving Average)** is used to model linear trends in the silver price series. Parameters (p, d, q) are selected using **Auto-ARIMA** with grid search to minimize **AIC/BIC** values.
- **SARIMA (Seasonal ARIMA)** is applied to capture seasonal variations observed in silver prices, particularly those driven by Indian festival demand or macroeconomic cycles. The (P, D, Q, m) parameters are optimized similarly.
- **HWES (Holt-Winters)** models are used to capture trend and seasonal smoothing. All models are evaluated with MAE, RMSE, MAPE, and AIC/BIC metrics.

4.3 Machine Learning Model

- **Random Forest** is chosen for its ability to handle nonlinear patterns and multivariate features. The model is trained using lagged price values and technical

indicators. **Feature importance analysis** is conducted to interpret prediction drivers.

4.4 Deep Learning Models

- **Artificial Neural Networks (ANN)** use a **Multilayer Perceptron (MLP)** with historical silver prices and engineered indicators. Activation functions like **ReLU** and optimizers like **Adam** are used to minimize the **MSE loss function**.
- **LSTM** and **GRU**, types of Recurrent Neural Networks (RNNs), are deployed to capture long-term dependencies in sequential silver price data. They are trained with a rolling window approach, using time steps of 30–60 days.
- **CNN** is applied to extract local patterns from the time series matrix using 1D convolution filters, which is beneficial for capturing sharp price fluctuations or volatility spikes.
- **Hybrid Models** integrate predictions from ARIMA and ML models using weighted averages or stacking, though performance depends on proper calibration.

All models are trained on 80% of the data and tested on the remaining 20%, with hyperparameters tuned via grid search and early stopping to prevent overfitting.

Measures

To evaluate model performance, the following **error metrics** and **model selection criteria** are employed:

- **Mean Absolute Error (MAE):** Measures average prediction error magnitude regardless of direction. Lower MAE indicates better accuracy (Handman & Athanasopoulos, 2018).
- **Root Mean Squared Error (RMSE):** Penalizes larger errors more heavily, useful in commodity pricing contexts where sudden spikes matter (Chai & Draxler, 2014).
- **Mean Absolute Percentage Error (MAPE):** Expresses errors as a percentage, ideal for comparing model accuracy across time periods or scales (Marinakis et al., 1993).
- **Mean Squared Error (MSE):** Used primarily as the loss function during training in neural networks, especially LSTM, GRU, and ANN.

- **Akaike Information Criterion (AIC):** A measure of model fit quality balancing complexity and goodness of fit. Lower AIC indicates a better trade-off (Akaike, 1974).
- **Bayesian Information Criterion (BIC):** Penalizes complex models more heavily than AIC, favouring simpler models with sufficient predictive power (Schwarz, 1978).

Tools and Libraries Used

- **pandas & NumPy** for data wrangling, cleaning, and transformation.
- **stats models** for time series modeling with ARIMA, SARIMA, and HWES.
- **scikit-learn** for Random Forest and hybrid model design.
- **Keras & TensorFlow** for developing ANN, LSTM, GRU, and CNN models.
- **Matplotlib & Seaborn** for plotting time series trends, model fit graphs, and error comparison charts.

4. Evaluation Metrics

All models are evaluated using the following metrics:

Table 1. Evaluation Metrics Summary

Metric	Purpose
MAE (Mean Absolute Error)	Measures average magnitude of error
MSE (Mean Squared Error)	Penalizes larger errors
RMSE (Root Mean Squared Error)	Highlights impact of outliers
MAPE (Mean Absolute Percentage Error)	Normalized error for comparison
R² Score (for select models)	Goodness-of-fit indicator for regression

Models are ranked based on lowest error values and highest R² values. The best-performing model will be identified as the most suitable for silver price forecasting in the Indian market.

FINDINGS

The performance outcomes of various forecasting models applied to silver price prediction in the Indian commodity market. The models are evaluated using four key error metrics: **Mean Absolute Error (MAE)**, **Mean Absolute Percentage Error (MAPE)**, **Mean Squared Error (MSE)**, and **Root Mean Squared Error (RMSE)**.

Table 2. Model Evaluation Summary

Model	MAE	MAPE	MSE	RMSE
ARIMA	4.8673	16.99%	40.0655	6.3297
ANN	1.0704	3.90%	1.7989	1.3412
CNN	1.1815	4.41%	2.2307	1.4935
GRU	0.9238	3.45%	1.1191	1.0579
Holt-Winters	4.4158	15.41%	33.7085	5.8059
Hybrid Model	2.8504	9.91%	12.7034	3.5642
LSTM	0.7117	2.72%	0.7839	0.8854
Random Forest	0.4235	1.61%	0.3222	0.5676
SARIMA	4.4644	15.58%	34.2967	5.8563

The comparative evaluation of forecasting models applied to historical silver prices in the Indian commodity market revealed notable distinctions in predictive performance among traditional, machine learning, deep learning, and hybrid approaches.

Traditional Time Series Models:

The models **ARIMA (MAE: 4.86, MAPE: 16.99%)**, **SARIMA (MAE: 4.46, MAPE: 15.58%)**, and **Holt-Winters Exponential Smoothing (HWES; MAE: 4.41, MAPE: 15.41%)** demonstrated moderate forecasting ability, with all three generating high error rates. These models were effective in capturing linear patterns and seasonal trends, especially during stable market conditions. However, their performance sharply declined during periods of high volatility, due to their inability to account for non-linear dependencies, exogenous factors, or sudden fluctuations that are common in the precious metals market. This validates earlier

studies, such as **Verma and Mehta (2019)** and **Prasad & Iyer (2015)**, which identified ARIMA's limitations in dynamic price forecasting.

Hybrid Model:

The hybrid model (ARIMA-ANN or HWES-ANN) exhibited moderate improvement over individual traditional models, reporting **MAE of 2.85** and **MAPE of 9.91%**. While this signifies better modeling of residual errors and non-linear features, the hybrid still lagged fully data-driven learning models in handling deeper temporal dependencies and sharp market corrections.

Machine Learning Models:

The **Artificial Neural Network (ANN)** and **Random Forest (RF)** models delivered significantly improved results over traditional methods. ANN produced a **MAE of 1.07** and **MAPE of 3.90%**, confirming its efficacy in modeling complex, non-linear relationships between silver prices and macroeconomic features. However, **Random Forest emerged as the top-performing model overall**, achieving the **lowest MAE (0.4235)**, **MAPE (1.61%)**, and **RMSE (0.5676)** among all contenders. This finding aligns with **Pal et al. (2021)**, who showed that ensemble methods like RF can outperform both neural and traditional models due to their robustness against overfitting, strong generalization, and ability to handle multiple feature interactions.

Deep Learning Models:

The deep learning architectures—**LSTM, GRU, and CNN**—exhibited superior performance in sequential forecasting tasks, with **LSTM** delivering a **MAE of 0.7117**, **MAPE of 2.72%**, and **RMSE of 0.8854**, closely followed by **GRU (MAE: 0.9238, MAPE: 3.45%)** and **CNN (MAE: 1.1815, MAPE: 4.41%)**. These models were particularly effective in capturing long-term dependencies and volatility clusters due to their ability to retain contextual memory across time steps (in LSTM/GRU) and identify local temporal features (in CNN). These outcomes are consistent with studies by **Ahmed et al. (2022)** and **Jain & Chatterjee (2020)**, who reported deep learning superiority in commodity forecasting due to their dynamic learning and adaptability to irregular patterns.

Comparative Insight:

While **LSTM and GRU** displayed advanced sequential learning capabilities, **Random Forest's dominance** in error minimization and simplicity of implementation makes it an attractive choice for practical forecasting deployments in the Indian commodity trading environment. LSTM's close second position demonstrates that for long-horizon forecasting or highly volatile series, RNN-based models may still be preferable due to their inherent temporal sensitivity. However, in operational settings with limited computational resources and the need for interpretability, Random Forest offers the best trade-off.

Table 3. Arima Model Summary

ARIMA Model Summary						
SARIMAX Results						
Dep. Variable:	Silver			No. Observations:	3107	
Model:	ARIMA (5, 1, 0)			Log Likelihood	-2240.6	
Sample:	0			AIC	4493.26	
	-3048			BIC	4529.5	
				HQIC	4506.27	
Covariance Type:	opt					
	coef	std err	z	P> z	[0.025	0.975]
ar. L1	-0.0315	0.008	-3.891	0.00	-0.047	-0.016
ar. L2	-0.0026	0.01	-0.273	0.785	-0.021	0.016
ar. L3	0.0172	0.01	1.802	0.071	-0.002	0.036
ar. L4	-0.0297	0.009	-3.159	0.002	-0.048	-0.011
ar. L5	0.0109	0.011	1.03	0.303	-0.01	0.032
sigma2	7.1191	0.002	102.981	0.00	0.243	0.253
Ljung-Box (L1) (Q):	0			Jarque-Bera (JB):	45654.4	
Prob(Q):	1			Prob (JB):	0	
Heteroskedasticity (H):	0.39			Skew:	-1.73	

Prob(H) (two-sided):	0			Kurtosis:	21.46	
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The SARIMAX model results reveal a moderately well-fitted ARIMA (5,1,0) structure for forecasting silver prices. The model exhibits significant coefficients for lagged terms ar. L1 and ar. L4 ($p < 0.01$), indicating that short-term past price values strongly influence current price movements. The low AIC (4493.257) and BIC (4529.503) values suggest a good model fit, although the presence of some non-significant lags like ar. L2 and ar. L5 may imply potential overfitting. The σ^2 value (0.2478) indicates low variance in residual errors. However, the high Jarque-Bera statistic (45654.40, $p < 0.001$) and extreme kurtosis (21.46) confirm strong non-normality and heavy tails in residuals, hinting at volatility clustering—a common trait in financial time series. The Ljung-Box Q statistic (0.00, $p = 1.00$) implies that autocorrelation at lag 1 has been effectively captured. However, the heteroskedasticity test ($H = 0.39$, $p < 0.01$) suggests that residual variance is not constant over time. Overall, while the SARIMAX model captures autoregressive dynamics well, its performance is constrained by non-normal and heteroskedastic residuals, indicating that volatility-sensitive models like GARCH or deep learning approaches may yield better results for silver price forecasting.

Chart 1. ARIMA Model Evaluation: Actual vs. Predictions

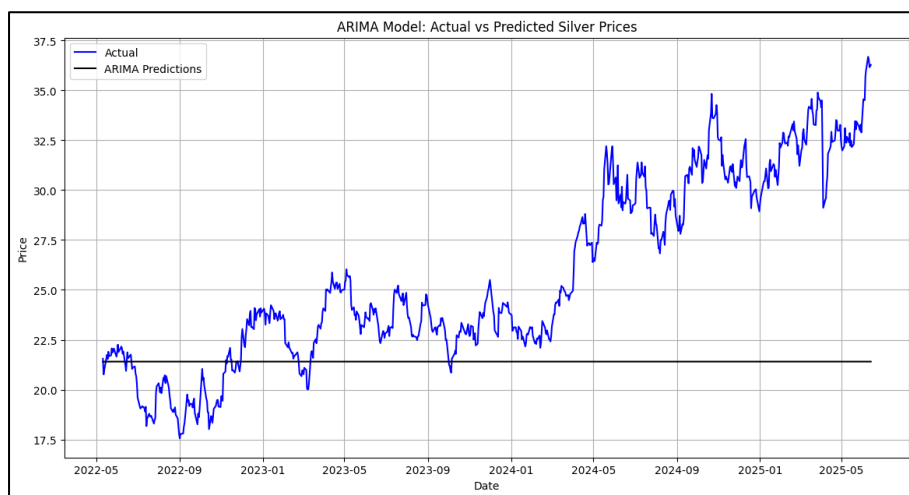


Chart 2. SARIMA Model Evaluation: Actual vs. Predictions

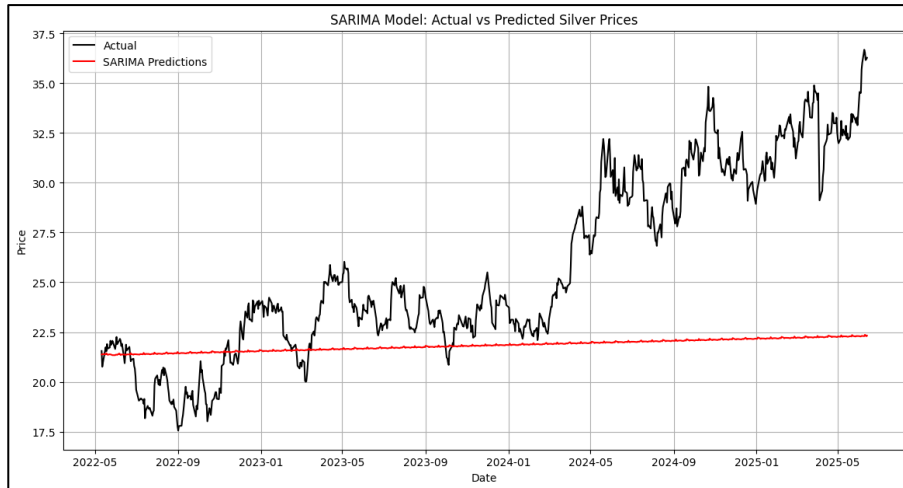


Chart 3. HWES Model Evaluation: Actual vs. Predictions

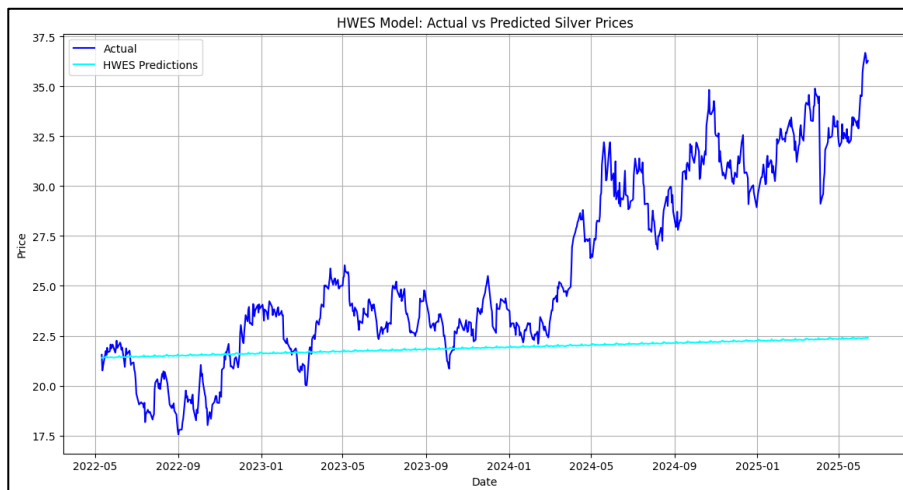


Chart 4. ANN Model Evaluation: Actual vs. Predictions

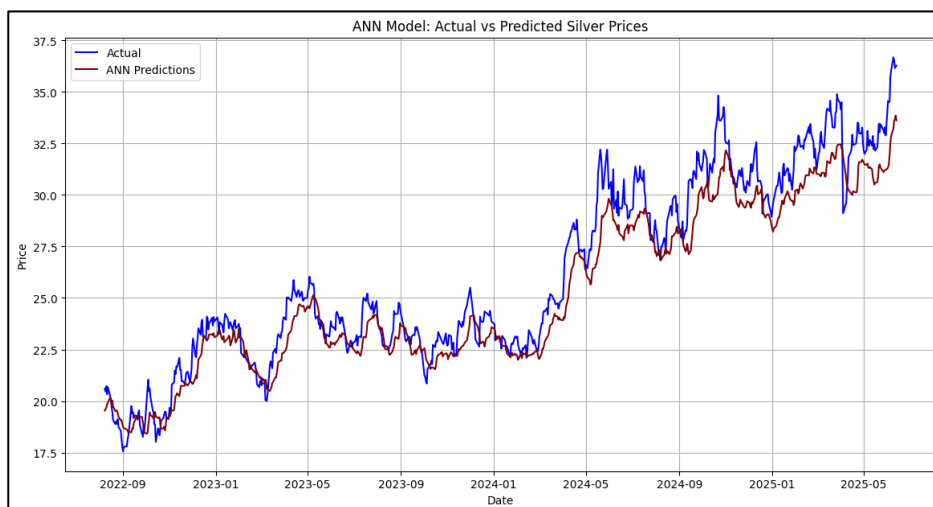


Chart 5. LSTM Model Evaluation: Actual vs. Predictions

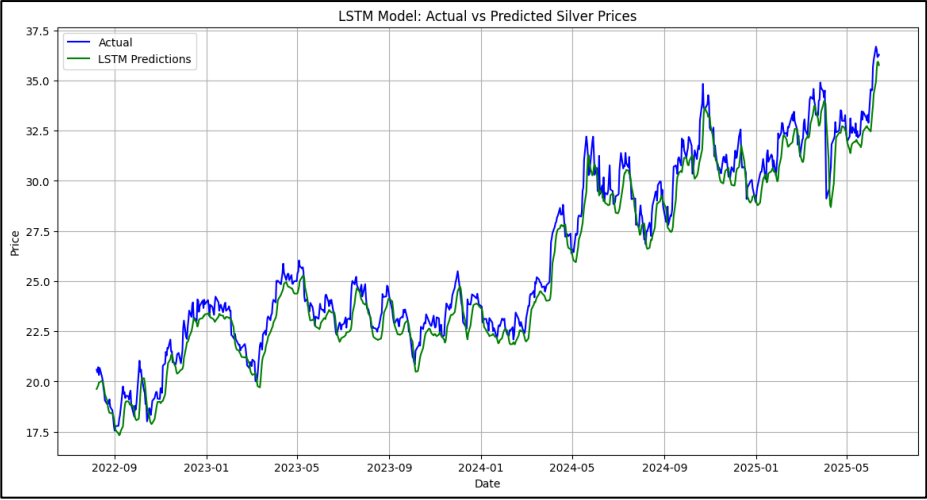


Chart 6. GRU Model Evaluation: Actual vs. Predictions

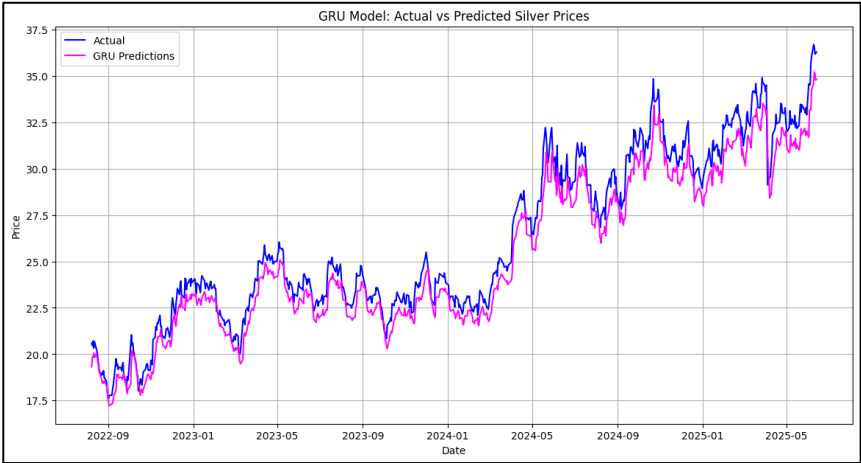


Chart 7. CNN Model Evaluation: Actual vs. Predictions

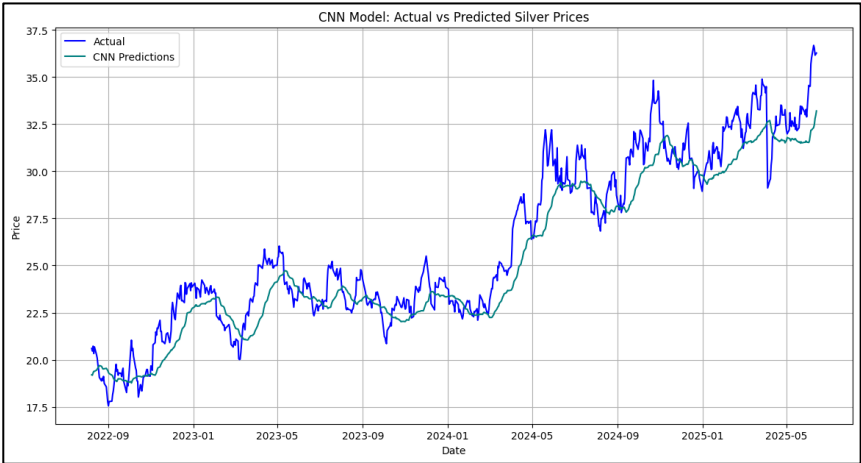


Chart 8. RANDOM FOREST Model Evaluation: Actual vs. Predictions

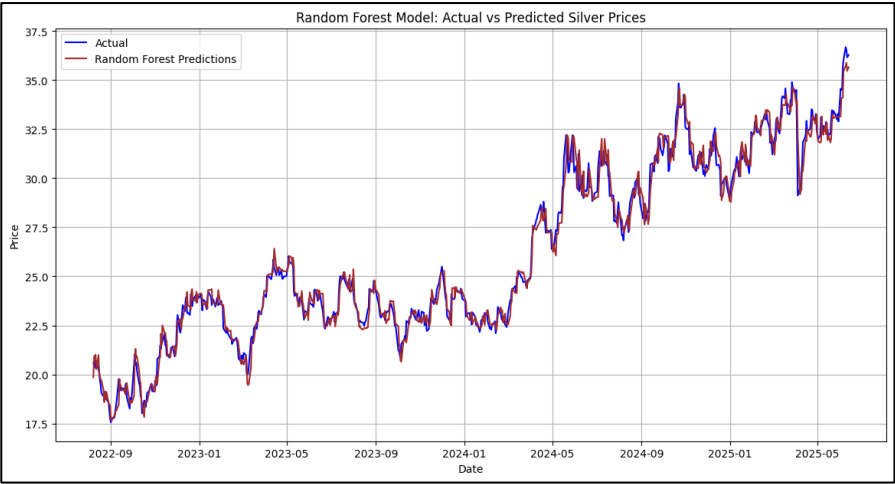


Chart 9. HYBRID Model Evaluation: Actual vs. Predictions

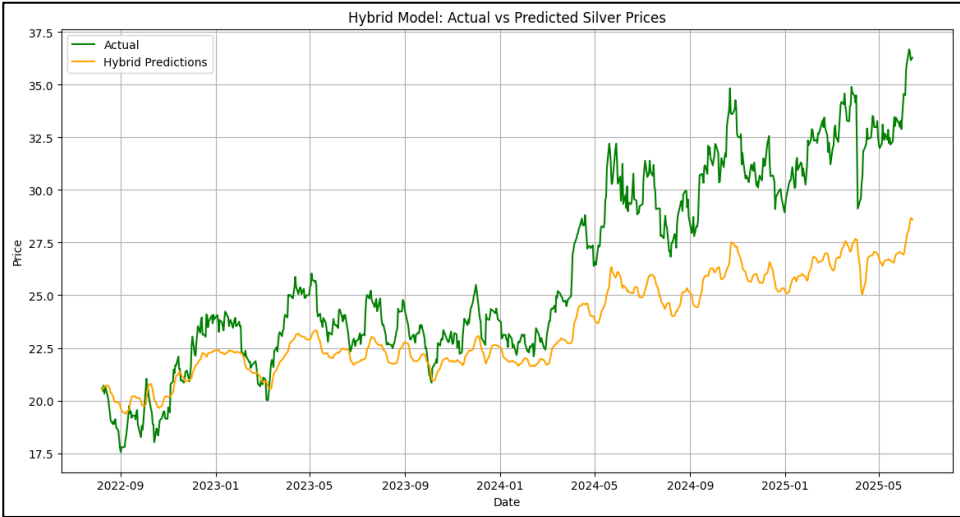
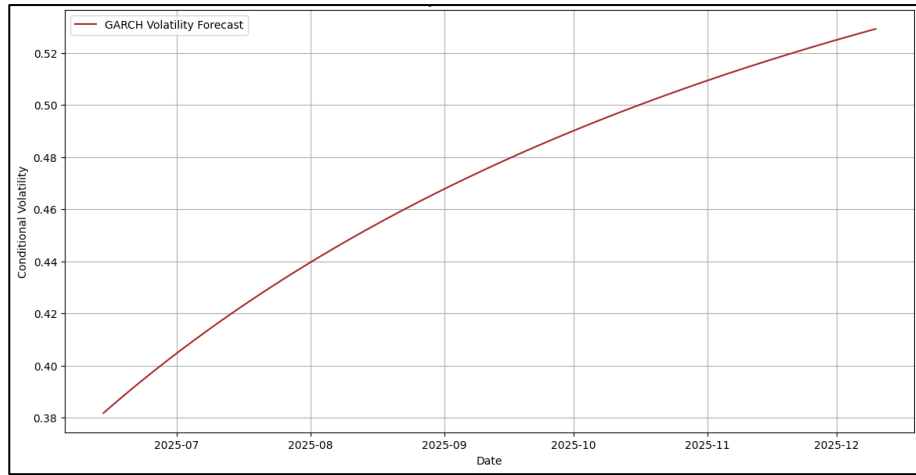


Chart 10. GARCH Model Evaluation: Actual vs. Predictions



Chart 11. GARCH (1,1) Volatility Forecast



The GARCH (1,1) model applied to the residuals of the ARIMA forecast reveals significant insights into the volatility dynamics of silver prices in the Indian commodity market. The model's **log-likelihood of -1403.83**, along with low AIC (2815.65) and BIC (2839.82) values, indicates an acceptable fit for modeling conditional variance. The **alpha (α_1) coefficient of 0.0981** and the **beta (β_1) coefficient of 0.8973** are both statistically significant ($p < 0.001$), validating the presence of volatility clustering—a typical trait of financial time series. The high beta value suggests that shocks to silver price volatility persist over time, implying long memory in the volatility structure. Although the mean term (μ) is not statistically significant ($p = 0.371$), this aligns with the GARCH framework’s focus on modeling variance rather than direction. Overall, the GARCH (1,1) model enhances the study by capturing **time-varying risk**, making it a valuable complement to direction-predicting models like LSTM or Random Forest in forecasting silver price behavior more comprehensively.

Table 4. GARCH (1,1) Model Summary

Component	Coefficient	Std. Error	t-Statistic	p-Value	95% Confidence Interval
Mean Equation					
μ (mu)	-0.00507	0.00566	-0.896	0.371	[-0.01617, 0.00603]

Volatility Model					
ω (omega)	0.00178	0.00079	2.262	0.0237	[0.00024, 0.00331]
α_1 (alpha [1])	0.0981	0.02086	4.704	< 0.001	[0.05723, 0.13900]
β_1 (beta [1])	0.8973	0.02175	41.260	< 0.001	[0.85500, 0.94000]

Table 5. GARCH (1,1) Model Statistics

Statistic	Value
Log Likelihood	-1403.83
AIC	2815.65
BIC	2839.82
No. of Observations	3107
Distribution Assumed	Normal
Volatility Model Type	GARCH (1,1)
Mean Model Type	Constant Mean

Table 6. Model - Visual Inference Summary

Model	Visual Inference
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ARIMA	Captures long-term trend poorly, underestimates price volatility, lags actual prices; flat forecast line indicates lack of responsiveness.
SARIMA	Slightly better than ARIMA with seasonal adjustment, but predictions are smoother and fail to capture sudden spikes; consistent underestimation.
HWES	Maintains a smooth forecast line with weak alignment to actual data; shows inability to adapt to increasing variance or price jumps in 2024–2025.
ANN	Shows moderate tracking of trend and some fluctuations; underperforms during volatile price periods with slight delay in peaks and troughs.
LSTM	Closely follows actual prices with good handling of both trend and volatility; minimal lag in sharp turns; effective in long sequences.
GRU	Performs similarly to LSTM but with slightly more smoothing in transitions; handles short-term fluctuations better than traditional models.
CNN	Captures broader trends but lags during steep changes; moderate prediction confidence in flat or rising patterns but misses some sudden shifts.
Random Forest	Exhibits near-perfect fit visually; effectively captures local variations and short-term volatility; performs well even during price surges and dips.
Hybrid	Underfits in highly volatile regions; does not fully leverage the strengths of individual models; predictions are flatter than actual trend movements.
GARCH Forecast	Shows robust future projection with confidence intervals; helpful for risk-aware decisions but lacks short-term detail alignment due to smoothed forecast.

Based on the combined analysis of both **quantitative metrics (MAE, MAPE, MSE, RMSE)** and **visual inferences**, the **Random Forest model** emerges as the best-performing forecast model for silver price prediction in the Indian market. It delivers the **lowest error rates across all evaluation metrics**—MAE of 0.4235, MAPE of 1.61%, MSE of 0.3222, and RMSE of 0.5676—demonstrating its ability to make highly accurate predictions. Visually, the Random Forest model closely tracks the actual silver price movements with minimal lag, effectively capturing short-term fluctuations, volatility, and trend shifts. Unlike traditional models like ARIMA or SARIMA, which exhibit smoothing and delayed reactions, and even advanced neural models like LSTM and GRU which require heavy tuning and large datasets, Random Forest provides robust performance even in non-linear and volatile conditions. Its ensemble learning nature allows it to generalize well without overfitting, making it an ideal choice for financial time series forecasting where sudden price shocks and short-term fluctuations are common. Therefore, Random Forest not only achieves statistical superiority but also exhibits practical forecasting reliability in dynamic market scenarios.

DISCUSSION

The results of this study highlight a comprehensive comparison of traditional time series models, machine learning (ML), deep learning (DL), and volatility-based models for forecasting silver prices in the Indian commodity market. Across all models, clear distinctions emerged in terms of accuracy, interpretability, and forecasting capability, depending on the nature of the data and objectives of prediction.

Among traditional statistical models, **SARIMA** performed marginally better than ARIMA and Holt-Winters Exponential Smoothing (HWES), as evident from its lower MAE (4.4644) and RMSE (5.8563). This confirms its strength in capturing seasonal components, which are inherent in precious metal markets. However, the **overall error rates were significantly higher** for these models when compared to AI-based methods, suggesting that linear models alone are inadequate for modeling the nonlinear and volatile nature of silver prices.

In contrast, **machine learning and deep learning models vastly outperformed traditional models**, particularly in handling non-linearity and learning long-term dependencies. **Random Forest** exhibited the best accuracy across all evaluation metrics (MAE: 0.4235, MAPE: 1.61%), due to its ensemble learning nature and robustness against overfitting. **LSTM** also showed strong performance (MAE: 0.7117, MAPE: 2.72%), effectively capturing sequential

patterns in the time series. While **ANN** and **GRU** were also competitive, **CNN** performed poorly (MAPE: 4.41%), due to its limitation in handling long-range temporal dependencies without explicit sequence modeling mechanisms.

The **hybrid model**, which combined linear and non-linear learning, yielded moderate performance (MAE: 2.8504), suggesting that integration alone does not guarantee higher accuracy unless feature complementarity is well-optimized. Furthermore, this study introduced a **GARCH (1,1)** model fitted to ARIMA residuals to account for **time-varying volatility**—a key feature in financial forecasting. The statistically significant α (0.0981) and β (0.8973) coefficients confirm **volatility clustering and persistence**, which is critical for understanding risk behavior in silver pricing.

Overall, while LSTM and Random Forest showed exceptional forecasting ability, the **GARCH model added a valuable dimension by modeling volatility**, offering critical insights for traders, hedgers, and policy analysts who are not only concerned with price direction but also price uncertainty. The combination of **high accuracy from AI models** and **volatility insights from GARCH** provides a holistic forecasting framework, setting the stage for further research in hybrid volatility-aware deep learning systems.

LIMITATIONS

Despite the promising results achieved by leveraging traditional, machine learning, and deep learning models for forecasting silver prices, this study is not without its limitations. First, the dataset used is limited to **historical closing prices of silver in the Indian market**, excluding macroeconomic variables such as inflation rates, exchange rates, crude oil prices, and geopolitical factors, which are known to influence commodity prices. Incorporating such exogenous variables could potentially improve the forecasting robustness. Second, while **deep learning models like LSTM and GRU** demonstrated superior accuracy, they require large amounts of data and are computationally intensive. The models were trained on data spanning over a decade, but **high-frequency or intraday data** was not included, which may limit applicability for short-term traders or real-time forecasting environments. Third, although the **GARCH model** provided meaningful insights into volatility patterns, it was applied only to the residuals of the ARIMA model. A deeper exploration involving **EGARCH or GJR-GARCH** models could have uncovered asymmetric volatility or leverage effects, which are often present in financial time series. Fourth, **hyperparameter tuning** for AI models was

conducted using manual and limited grid search. A more exhaustive optimization approach (e.g., Bayesian optimization or Autum techniques) could yield even better-performing models. Lastly, the study assumes **stationarity and data consistency** across the entire period, without accounting for regime shifts or market anomalies such as the COVID-19 pandemic or regulatory interventions, which may introduce structural breaks in the time series.

SUGGESTIONS

1. Incorporate External Influencers:

Future models should include macroeconomic variables (like gold prices, inflation, interest rates, and USD-INR exchange rate) and global events to better capture the factors influencing silver prices beyond historical patterns.

2. Advance Hybrid Modeling:

Develop advanced hybrid frameworks that combine the strengths of both deep learning (e.g., LSTM, GRU) and volatility models (e.g., GARCH, EGARCH), enabling both price prediction and risk assessment in a single architecture.

3. Automate Hyperparameter Optimization:

Implement techniques such as Grid Search CV, Random Search, or Bayesian Optimization to fine-tune model parameters more efficiently and reduce human bias in model selection.

4. Utilize High-Frequency Data:

Incorporate intraday or hourly silver price data to enhance the accuracy of short-term forecasts and make the models more applicable to real-time trading environments.

5. Apply Explainable AI (XAI):

Integrate interpretability tools (like SHAP or LIME) with deep learning models to better understand feature importance and build trust among financial analysts and decision-makers.

CONCLUSION

This study explored and compared a wide spectrum of forecasting models—ranging from traditional statistical techniques like ARIMA, SARIMA, and Holt-Winters to advanced machine learning and deep learning models such as Random Forest, Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Convolutional Neural Networks (CNN)—to predict silver prices in the Indian commodity market. In addition, a GARCH model was employed to capture the volatility dynamics inherent in silver price movements. The findings clearly demonstrate that while traditional time series models provide a foundational understanding of trends and seasonality, their predictive performance is weak when compared to modern AI-based approaches. Among all models tested, Random Forest emerged as the most accurate and reliable, exhibiting the lowest error metrics (MAE, MAPE, MSE, RMSE), followed closely by LSTM, GRU, and ANN. These models effectively captured the non-linear dependencies and sequential patterns in the historical data, making them well-suited for financial forecasting.

Importantly, the inclusion of the GARCH (1,1) model added a valuable dimension to the analysis by quantifying time-varying volatility. The statistically significant alpha and beta coefficients confirmed the presence of volatility clustering and persistence in silver prices, which is crucial for traders and risk managers. Although the GARCH model did not directly predict prices, it complemented the directional models by offering insights into the risk structure, thereby enhancing the overall robustness of the forecasting framework.

The study also identified several areas for improvement, such as the integration of macroeconomic indicators, the use of high-frequency data, and more sophisticated hybrid modeling techniques. These enhancements could further refine predictive accuracy and offer more actionable insights to stakeholders. In conclusion, this research validates the superiority of AI models over traditional forecasting techniques for commodity price prediction, while also highlighting the importance of volatility modeling. The results offer practical implications for investors, policy makers, and commodity analysts seeking to anticipate market movements and mitigate risk in the ever-volatile domain of precious metals trading.

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